MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 11 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | $(4 \times 1 = 4$ Marks)

Q1. The perimeter of a rectangle is 44 cm and its length is 1 cm more than twice its breadth. The dimensions are: [1]

(A) 15 cm, 7 cm (B) 16 cm, 6 cm (C) 14 cm, 8 cm (D) 18 cm, 4 cm

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 15 cm, 7 cm** [0.5 Mark]

Justification: $2(l+b) = 44 \implies l+b = 22$, and $l = 2b+1 \implies 3b+1 = 22 \implies b = 7$ cm, $l = \mathbf{15}$ cm. [0.5 Mark]

Q2. If $x = 2, y = 3$ is a solution of the equation $5x - 3y = k$, then the value of k is: [1]

(A) 1 (B) -1 (C) 19 (D) 0

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 1** [0.5 Mark]

Justification: $k = 5(2) - 3(3) = 10 - 9 = \mathbf{1}$. [0.5 Mark]

Q3. If $2x + 3y = 11$ and $2x - 4y = -24$, then the value of m for which $y = mx + 3$ is: [1]

(A) 1 (B) -1 (C) 2 (D) -2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) -1** [0.5 Mark]

Justification: Subtracting gives $7y = 35 \implies y = 5, x = -2$. Sub in $5 = -2m + 3 \implies m = \mathbf{-1}$. [0.5 Mark]

Q4. The value of k for which the system $kx - y = 2$ and $6x - 2y = 3$ has a unique solution is: [1]

(A) $k = 3$ (B) $k \neq 3$ (C) $k = 0$ (D) $k \neq 0$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $k \neq 3$** [0.5 Mark]

Justification: Unique solution requires $\frac{k}{6} \neq \frac{-1}{-2} \implies \frac{k}{6} \neq \frac{1}{2} \implies \mathbf{k \neq 3}$. [0.5 Mark]

SECTION B: Assertion-Reason Questions MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. **Assertion (A):** The lines $x = 3$ and $y = -2$ are perpendicular to each other.
Reason (R): The line $x = 3$ is parallel to the y-axis and $y = -2$ is parallel to the x-axis, and the coordinate axes are perpendicular. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\text{(A)}}$ [0.5 Mark]

Verification: Since $x = 3 \parallel y$-axis, $y = -2 \parallel x$-axis, and x-axis $\perp$ y-axis, both statements are true and (R) explains (A). [0.5 Mark]

Q6. **Assertion (A):** The equations $2x + 3y = 5$ and $4x + 6y = 15$ represent parallel lines.
Reason (R): Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\text{(A)}}$ [0.5 Mark]

Verification: $\frac{2}{4} = \frac{3}{6} = \frac{1}{2} \neq \frac{-5}{-15} = \frac{1}{3}$. Both (A) and (R) are true and (R) explains (A). [0.5 Mark]

SECTION C: Very Short Answer Questions MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. Find the value of k for which the system $2x + 3y = 5$ and $4x + ky = 10$ has infinitely many solutions. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Ratio Condition [2 Marks]

$\dfrac{2}{4} = \dfrac{3}{k} = \dfrac{5}{10} \implies \dfrac{1}{2} = \dfrac{3}{k} \implies \boxed{\mathbf{k = 6}}$.

Q8. Solve using elimination: $47x + 31y = 63$ and $31x + 47y = 15$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Adding & Subtracting Equations [1 Mark]

Adding: $78(x + y) = 78 \implies x + y = 1$ — (i) Subtracting: $16(x - y) = 48 \implies x - y = 3$ — (ii)

Step 2: Solving Simple Pair [1 Mark]

Adding (i) and (ii): $2x = 4 \implies x = 2$. Then $y = 1 - 2 = -1$. $\boxed{\mathbf{x = 2,\quad y = -1}}$

Q9. Write a pair of linear equations which has the unique solution $x = -1, y = 3$. How many such pairs exist? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Constructing Two Non-Parallel Lines Through $(-1, 3)$ [1 Mark]

Any $ax + by = a(-1) + b(3)$ passes through $(-1, 3)$. Choosing $(1, 1)$ and $(1, -1)$: $x + y = 2$ and $x - y = -4$ (check $\frac{1}{1} \neq \frac{1}{-1}$).

Step 2: Count of Such Pairs [1 Mark]

Since infinitely many distinct lines pass through $(-1, 3)$, **infinitely many** such pairs exist.

$\boxed{\text{Example: } \mathbf{x + y = 2},\ \mathbf{x - y = -4};\quad \text{Infinitely many pairs}}$

Q10. Four years ago, a mother was four times as old as her daughter. Six years later, the mother will be two and a half times as old as her daughter. Find their present ages. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Framing Equations [1.5 Marks]

Let mother's present age $= x$, daughter's $= y$.

4 years ago: $x - 4 = 4(y - 4) \implies x - 4y = -12$ — (i)

6 years later: $x + 6 = \frac{5}{2}(y + 6) \implies 2x + 12 = 5y + 30 \implies 2x - 5y = 18$ — (ii)

Step 2: Solving for y and x [1.5 Marks]

Substituting $x = 4y - 12$ into (ii): $2(4y - 12) - 5y = 18 \implies 3y = 42 \implies y = 14$ years.

Then $x = 4(14) - 12 = 56 - 12 = 44$ years. Mother = **44** years, Daughter = **14** years

Q11. Solve the following pair of linear equations graphically:

$$2x + 3y = 12 \quad \text{and} \quad x - y = 1$$

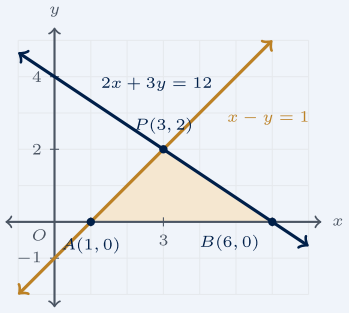
Find the coordinates of the vertices of the triangle formed by these lines and the line $y = 0$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Solution Tables [1 Mark]

x	0	3	6
$y = \frac{12-2x}{3}$	4	2	0

x	0	1	3
$y = x - 1$	−1	0	2



Step 2: Unique Graphical Solution [1 Mark]

Both lines intersect at $P(3, 2)$, giving $\mathbf{x = 3}$, $\mathbf{y = 2}$.

Step 3: Vertices with Line $y = 0$ [1 Mark]

The lines intersect $y = 0$ (x-axis) at $A(1, 0)$ and $B(6, 0)$. Hence $\triangle PAB$ has vertices $(1, 0), (6, 0), (3, 2)$ (Base = 5, Height = 2, Area = 5 sq. units).

Solution: $(\mathbf{3}, \mathbf{2})$; Triangle Vertices: $(\mathbf{1}, \mathbf{0})$, $(\mathbf{6}, \mathbf{0})$, $(\mathbf{3}, \mathbf{2})$

Q12. If the pair of linear equations $(k + 1)x + 3ky + 15 = 0$ and $5x + ky + 5 = 0$ represent coincident lines, find the value of k. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Formula / Theorem Used: Coincident Lines: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Step 1: Equating Ratios [1.5 Marks]

$$\frac{k + 1}{5} = \frac{3k}{k} = \frac{15}{5} = 3 \qquad (k \neq 0)$$

Step 2: Solving & Checking $k \neq 0$ [1.5 Marks]

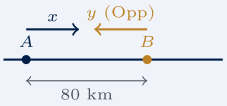
$\frac{k+1}{5} = 3 \implies k + 1 = 15 \implies k = 14$. (Checking $k = 0$ gives $x + 15 = 0$ and $5x + 5 = 0$, parallel lines, not coincident.)

$\mathbf{k = 14}$

Q13. Points A and B are 80 km apart from each other on a national highway. A car starts from A and another car starts from B at the same time. If they move in the same direction, they meet in 8 hours, and if they move towards each other, they meet in 1 hour and 20 minutes. Find the constant speeds of the two cars. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Speed of Car $A = x$ km/h, Car $B = y$ km/h ($x > y$).
Opposite time $= 1$ h 20 min $= 1 + \frac{20}{60} = \frac{4}{3}$ h.



Formula / Theorem Used: Same Dir Speed $= x - y$, Opposite Speed $= x + y$

Step 1: Relative Speed Equations [2 Marks]

Same dir: $8(x - y) = 80 \implies x - y = 10$ — (i)

Opposite dir: $\frac{4}{3}(x + y) = 80 \implies x + y = \frac{80 \times 3}{4} = 60$ — (ii)

Step 2: Elimination & Conclusion [3 Marks]

Adding (i) and (ii): $2x = 70 \implies x = 35$ km/h. From (ii): $y = 60 - 35 = 25$ km/h.

Car $A = \mathbf{35}$ km/h, Car $B = \mathbf{25}$ km/h

Q14. CASE STUDY: Highway Surveillance & Dual Patrol Vehicle Tracking

Two highway police patrol stations, Post Alpha and Post Beta, are positioned along National Highway 44 at a distance of 100 km apart. Highway radar cameras detected two suspect vehicles, Car 1 starting from Post Alpha and Car 2 starting from Post Beta at the same instant. If both vehicles travel in the same direction along the highway at constant speeds, Car 1 catches up with Car 2 in 5 hours. If they travel towards each other, they pass each other in 1 hour. Let u km/h be the speed of Car 1 and v km/h be the speed of Car 2 ($u > v$). [4]

(i) Formulate the pair of linear equations in u and v representing the two traffic scenarios. [1]

(ii) Determine the speed of Car 1 (u). [1]

(iii) Determine the speed of Car 2 (v). [1]

(iv) Find the distance traveled by Car 1 before meeting Car 2 when they move towards each other. [1]

— OR —

Find the ratio of the speed of Car 1 to that of Car 2. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Relative Speed Equations [1 Mark]

Same direction: $5(u - v) = 100 \implies u - v = 20$ — (i) Opposite direction: $1(u + v) = 100 \implies u + v = 100$ — (ii)

Sub-part (ii): Speed of Car 1 (u) [1 Mark]

Adding (i) and (ii): $2u = 120 \implies \boxed{\mathbf{u = 60}\text{ km/h}}$.

Sub-part (iii): Speed of Car 2 (v) [1 Mark]

From (ii): $v = 100 - 60 = \boxed{\mathbf{v = 40}\text{ km/h}}$.

Sub-part (iv): Distance Traveled by Car 1 in Opposite Motion [1 Mark]

They meet after 1 hour, so Distance $= u \times t = 60 \times 1 = \boxed{\mathbf{60}\text{ km}}$.

Sub-part (iv) [OR Option]: Ratio of Speeds ($u : v$) [1 Mark]

$u : v = 60 : 40 = \boxed{\mathbf{3 : 2}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★